Cybersecurity Intern

1. Introduction

This internship focused on developing foundational knowledge and practical skills in the field of cybersecurity, with an emphasis on network security and web application security. The primary objective was to understand how modern systems are protected against common threats and how vulnerabilities can be identified using industry-relevant tools and techniques.

During the internship, key concepts such as network threats, secure configurations, and traffic analysis were explored. Additionally, hands-on experience was gained in identifying web application vulnerabilities, including SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF), using tools like WebGoat and OWASP ZAP.

The tasks were designed to provide both theoretical understanding and practical exposure, enabling the development of problem-solving skills and a deeper insight into real-world cybersecurity challenges. Overall, this internship served as a valuable step toward building a strong foundation in cybersecurity and ethical hacking practices.

2.1 Task 1: Introduction to Network Security Basics

1. Overview

The objective of this task was to understand the fundamentals of network security, identify common network threats, and implement basic security measures to protect a small network environment. Additionally, network traffic was monitored using Wireshark to analyze different types of communication and detect any unusual activity.

2. Network Security Concepts

Network security involves protecting computer networks from unauthorized access, misuse, or attacks. The following common threats were studied:

- **Virus:** A malicious program that attaches itself to files and spreads when those files are executed.
- **Worm:** A self-replicating malware that spreads across networks without user interaction.
- **Trojan:** A type of malware disguised as legitimate software that allows unauthorized access to the system.
- **Phishing:** A social engineering attack where attackers trick users into providing sensitive information such as passwords or credit card details.

Key security concepts include:

- **Firewall:** A security system that monitors and controls incoming and outgoing network traffic based on predefined rules.
- **Encryption:** The process of converting data into a secure format to prevent unauthorized access.
- **Secure Network Configuration:** Practices such as using strong passwords and enabling secure WiFi protocols (WPA2/WPA3).

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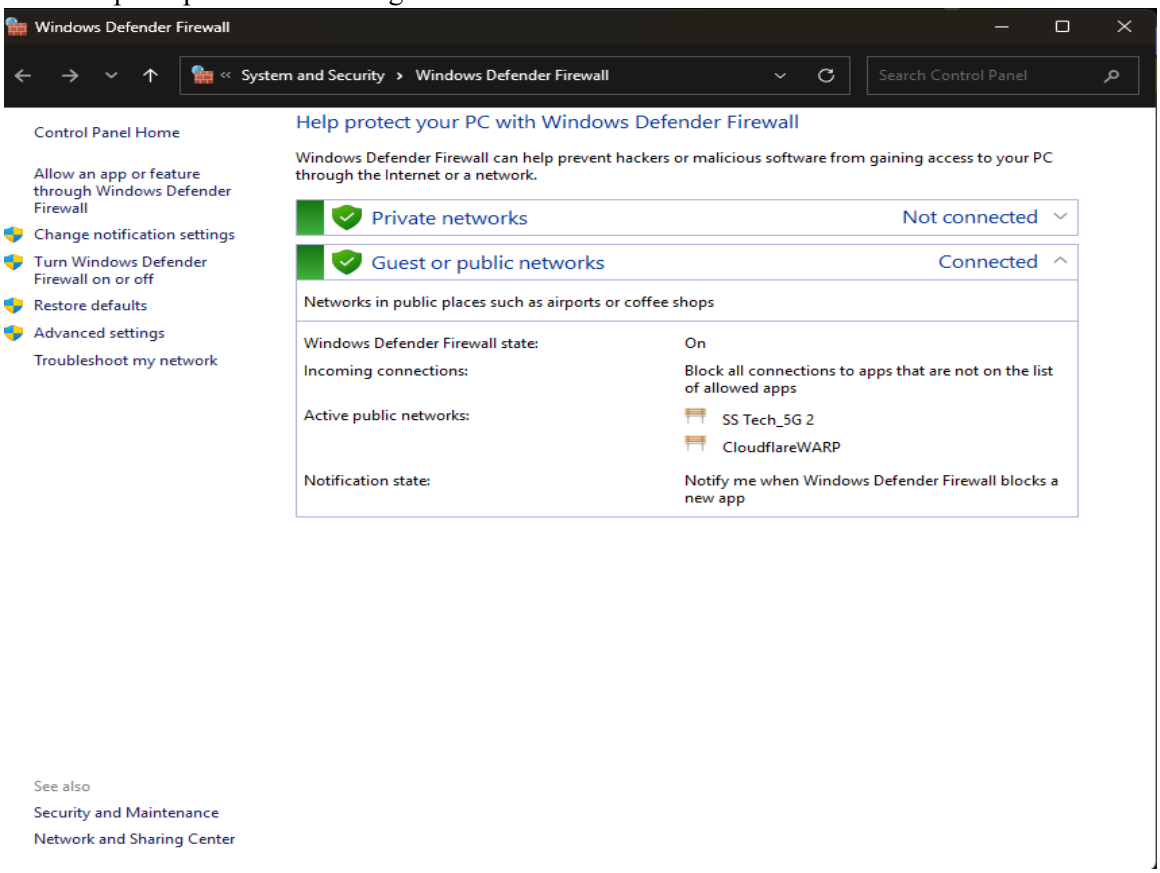
3. Implementation of Security Measures

To enhance network security, the following measures were implemented:

- Enabled **Windows Defender Firewall** for both private and public networks to block unauthorized access.
- Secured the WiFi network by ensuring a strong password and verifying the use of WPA2/WPA3 encryption.
- Reviewed basic system security settings to reduce vulnerabilities.

These steps helped in establishing a secure network environment.



4. Network Traffic Monitoring using Wireshark

Wireshark was used to capture and analyze live network traffic. The following observations were made:

- **DNS Traffic:** Used to resolve domain names (e.g., google.com) into IP addresses.
- **HTTP/HTTPS Traffic:** Represents web browsing activities.
- **TLS Protocol:** Indicates encrypted communication over secure websites.

Traffic was captured while browsing websites such as Google and YouTube. This helped in understanding how data flows through the network.

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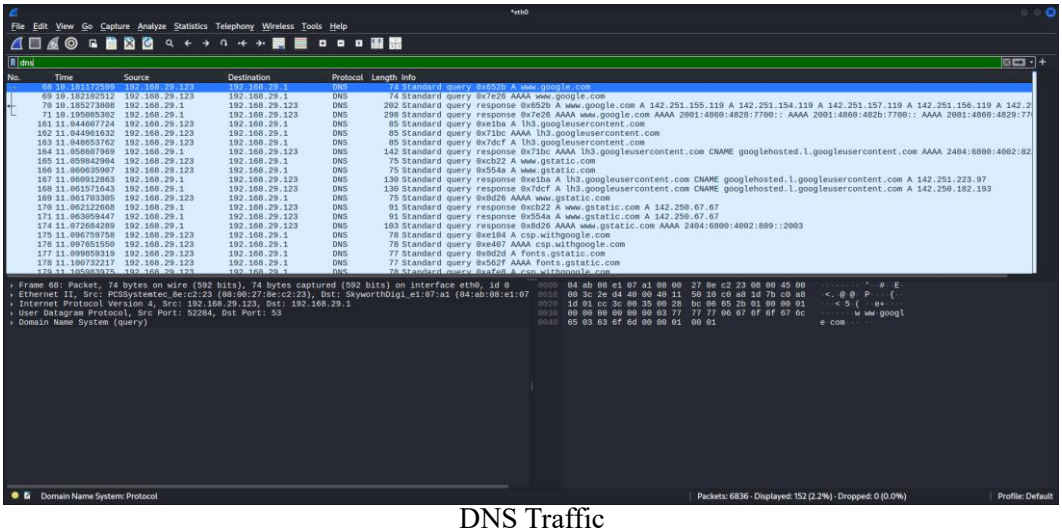
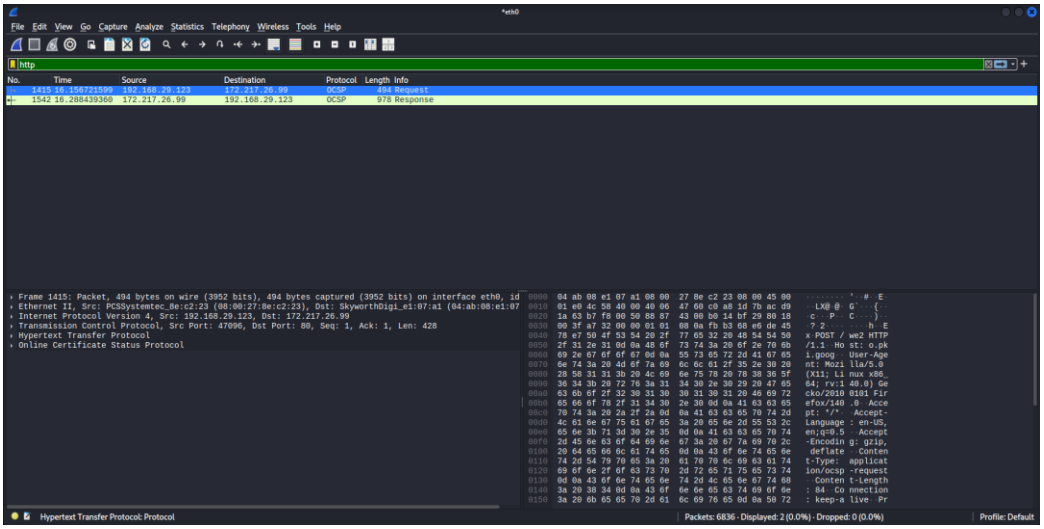
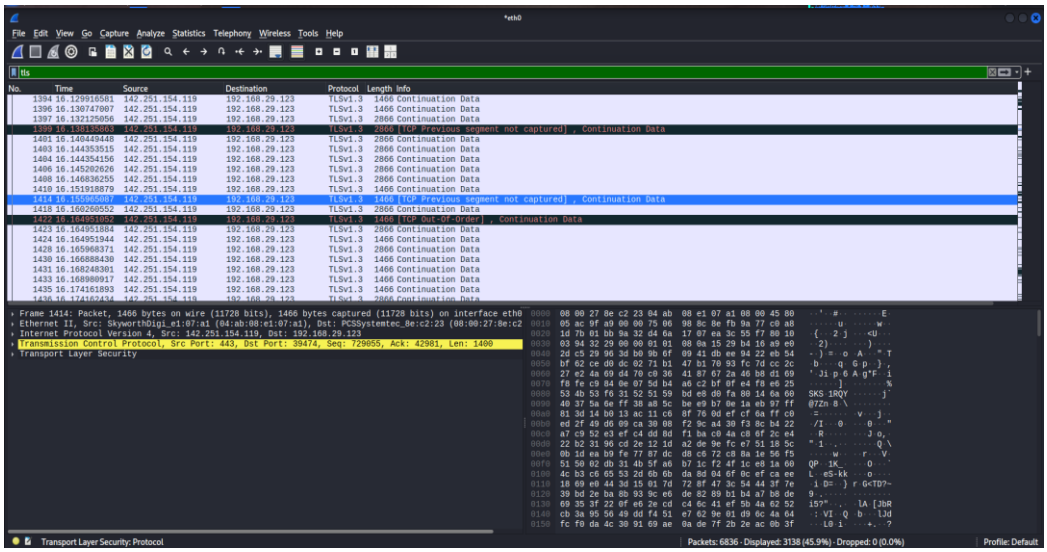


DNS Traffic



HTTP Traffic



HTTPS Traffic

5. Identification of Network Activity

During analysis, the captured traffic mainly showed normal user activity such as web browsing and DNS requests. No suspicious or malicious traffic was identified. However, unusual patterns such as unknown IP addresses or excessive repeated requests could indicate potential threats.

6. Importance of Security Measures

Basic security measures such as firewalls and encryption play a crucial role in protecting networks from attacks. They help in preventing unauthorized access, securing sensitive data, and maintaining system integrity.

7. Results

This task provided a foundational understanding of network security, common threats, and practical implementation of security measures. The use of Wireshark helped in visualizing real-time network communication, enhancing the understanding of how networks operate and how they can be protected.

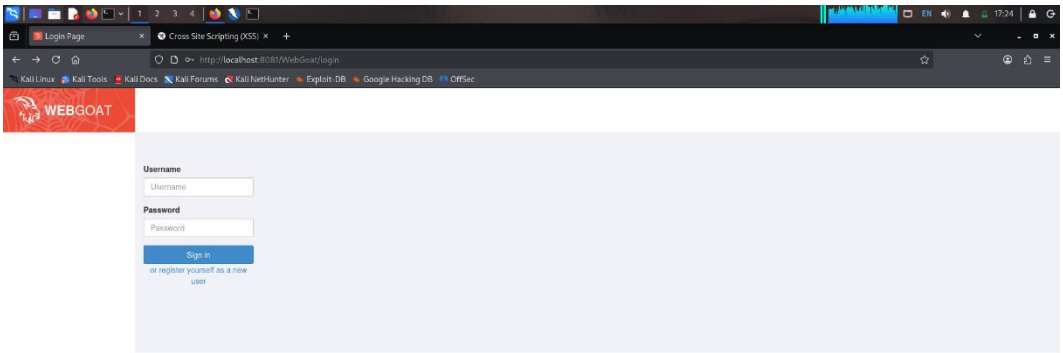
2.2 Task 2: Introduction to Web Application Security

1. Overview

The objective of this task was to understand common web application vulnerabilities and how they can be identified using security tools. A vulnerable web application was analyzed to gain practical knowledge of how attackers exploit weaknesses in web systems.

2. Tools and Methods Used

- **WebGoat:** Used as a deliberately vulnerable web application to practice and understand security flaws.



WebGoat

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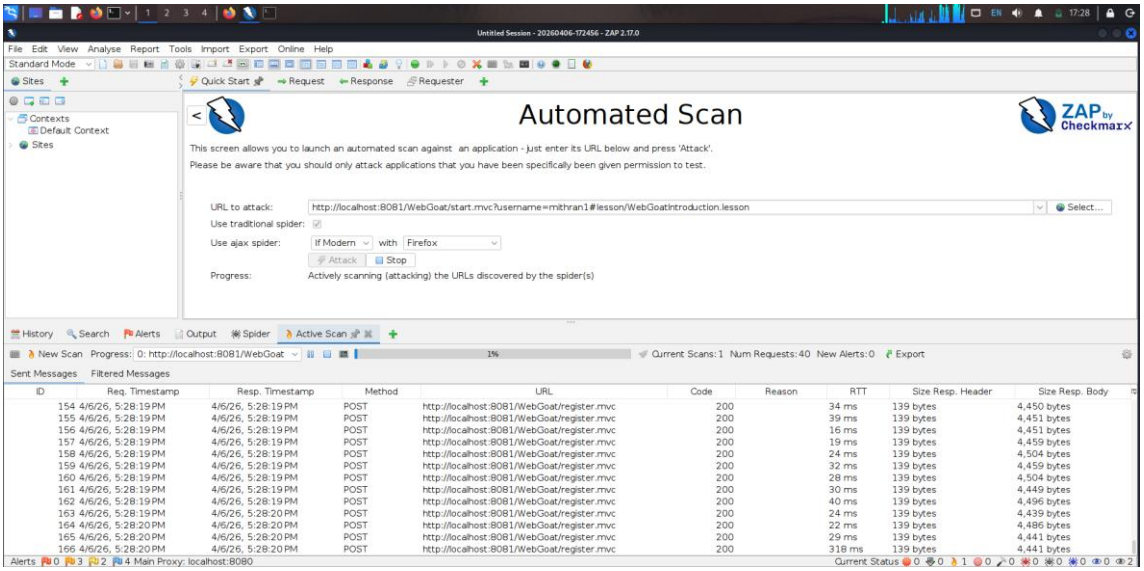
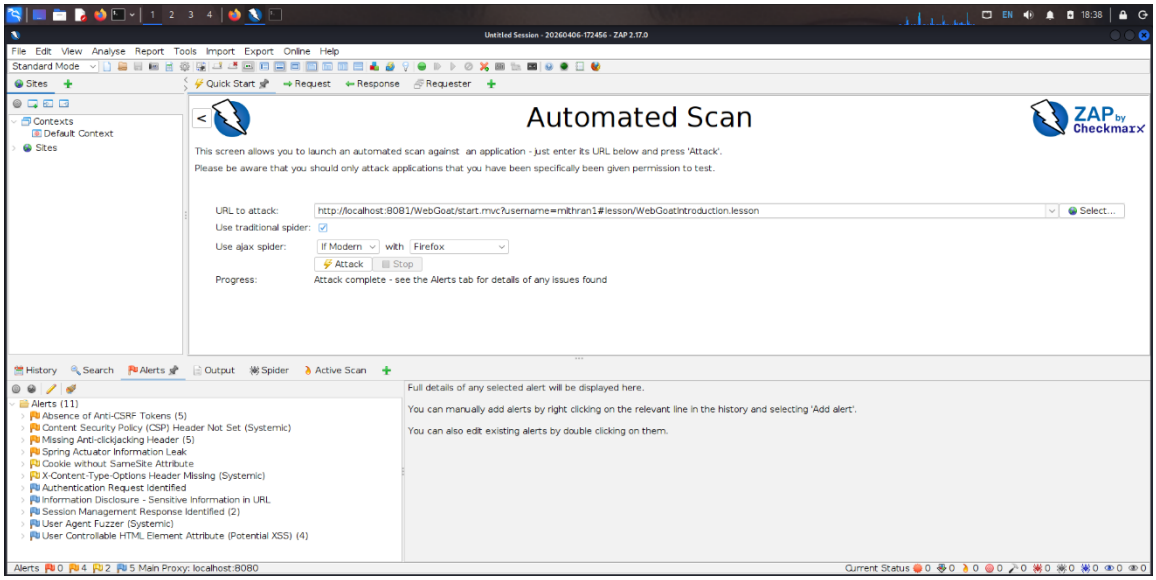
- **OWASP ZAP:** Used as a web application security scanner to identify vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF).



OWASP ZAP

The methodology included scanning the application using OWASP ZAP and performing manual testing on input fields to validate the identified vulnerabilities.



OWASP ZAP

3. Steps Taken to Complete the Task

- Set up the WebGoat application in a local environment using Docker.
- Accessed the application through the browser and registered a user account.
- Configured OWASP ZAP and initiated an automated scan on the WebGoat application.

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- Reviewed the alerts generated by ZAP to identify potential vulnerabilities.
- Performed manual testing:
  - Tested **SQL Injection** by entering SQL payloads in input fields.
  - Tested **XSS** by injecting JavaScript code into input fields.
- Observed the behavior of the application to confirm vulnerabilities.
- Captured screenshots as proof of findings.

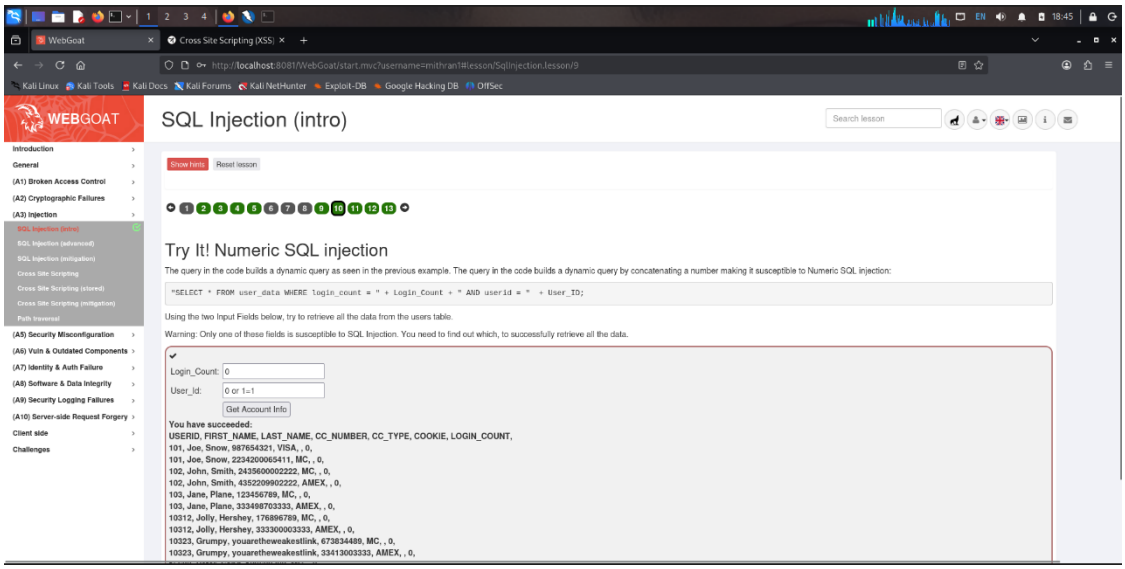
4. Challenges Faced and How They Were Overcome

- Initially faced issues with running the application due to port conflicts, which were resolved by using alternative ports.
- Difficulty in identifying the correct input fields for testing was overcome by exploring different sections of the application.
- Understanding vulnerability behavior required additional learning, which was addressed by reviewing OWASP documentation and tool descriptions.

5. Results and Outcomes

The following vulnerabilities were successfully identified:

- **SQL Injection:** It was possible to manipulate database queries by injecting SQL statements into input fields, demonstrating how attackers can access or modify database information.



WebGoat – SQL Injection

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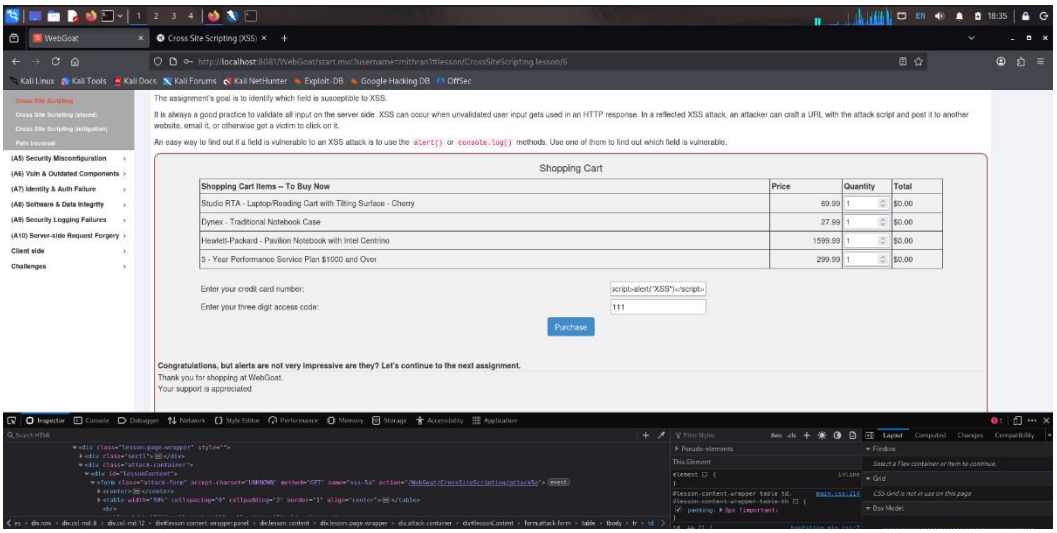
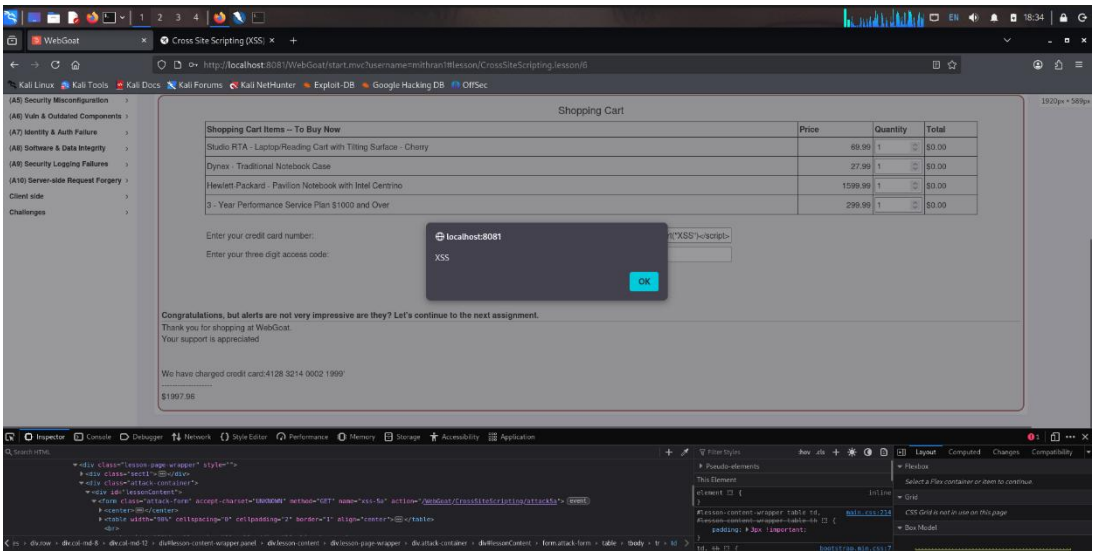
- **Cross-Site Scripting (XSS):** JavaScript code was successfully executed in the browser, indicating that the application did not properly validate user input.



Cross-Site Scripting (XSS)



Cross-Site Scripting (XSS) [Pop-up]

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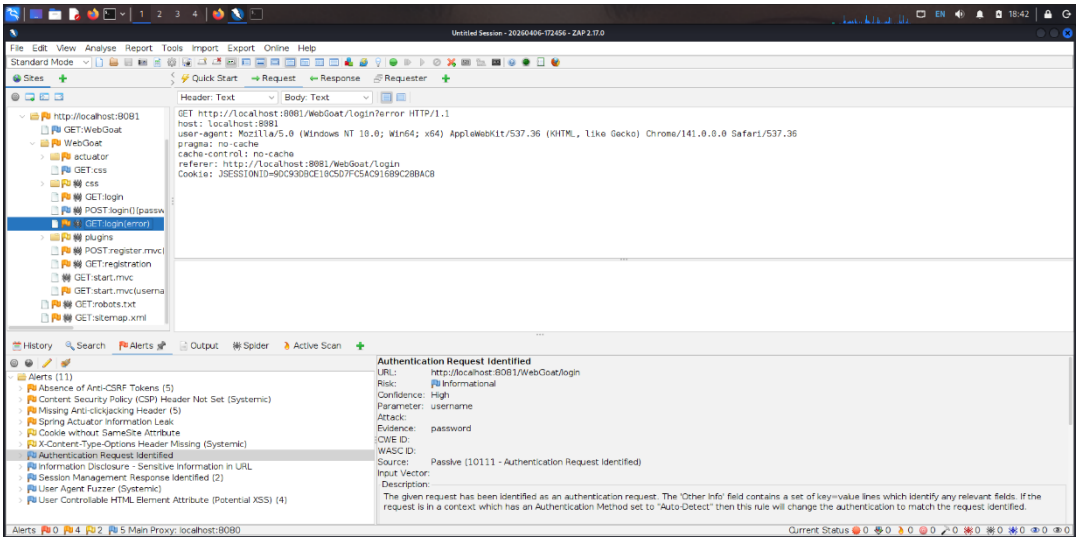
- **Cross-Site Request Forgery (CSRF):** Identified through OWASP ZAP alerts, showing that certain actions could be performed without proper user validation.



Cross-Site Request Forgery (CSRF)

Additionally, the use of OWASP ZAP helped in understanding automated vulnerability scanning and how security tools assist in identifying risks.

3. Conclusion

This internship task provided valuable hands-on experience in web application security. It enhanced my understanding of common vulnerabilities such as SQL Injection, XSS, and CSRF, and demonstrated how these issues can be identified using both automated tools and manual testing techniques. The practical exposure to tools like WebGoat and OWASP ZAP improved my ability to analyze and secure web applications, contributing to my overall growth in the cybersecurity domain.

4. Closing Remarks

I would like to express my gratitude for the opportunity to work on these tasks, which significantly improved my practical knowledge of cybersecurity concepts. The internship provided a strong foundation in identifying and understanding web application vulnerabilities. I look forward to applying these skills in real-world scenarios and continuing to enhance my expertise in the field of cybersecurity.